Choi matrix

We shall see some properties of Choi matrix.

Let $\varphi : M_m \to M_n$ be a hermitian preserving maps. Its Choi matrix is defined as

$$C_\varphi = \sum_{i,j=1}^{m} \varphi\big(E_{ij}\big) \otimes E_{ij} \in M_n \otimes M_m$$

Observe that

$$\langle E_{ij}, \varphi^*\big(F_{\alpha\beta}\big)\rangle = \langle \varphi\big(E_{ij}\big), F_{\alpha\beta}\rangle = \langle F_{\beta\alpha}, \varphi\big(E_{ji}\big)\rangle$$

$$\varphi^*\big(F_{\alpha\beta}\big) = \sum_{i,j=1}^{m} \langle F_{\beta\alpha}, \varphi\big(E_{ji}\big)\rangle) E_{ij}$$

Denote $OO^* := \sum_{\alpha,\beta=1}^{n} F_{\alpha\beta} \otimes F_{\alpha\beta}$. Given $X \in M_n \otimes M_m$.

$$(\mathrm{id}_n \otimes \varphi)^*(OO^*) = \sum_{\alpha,\beta=1}^{n} F_{\alpha\beta} \otimes \varphi^*\big(F_{\alpha\beta}\big) = \sum_{\alpha,\beta=1}^{n} \sum_{i,j=1}^{m} \langle F_{\beta\alpha}, \varphi\big(E_{ji}\big)\rangle F_{\alpha\beta} \otimes E_{ij} =$$

$$\sum_{i,j=1}^{m} \left(\sum_{\alpha,\beta=1}^{n} \langle F_{\beta\alpha}, \varphi\big(E_{ji}\big)\rangle F_{\alpha\beta} \right) \otimes E_{ij} = \sum_{i,j=1}^{m} \left(\sum_{\alpha,\beta=1}^{n} \langle F_{\beta\alpha}, \varphi\big(E_{ji}\big)\rangle F_{\beta}^{\top} \right) \otimes E_{ji}^{\top} = C_\varphi^{\top}$$

$$\langle (\mathrm{id}_n \otimes \varphi)(X), OO^*\rangle = \langle X, (\mathrm{id}_n \otimes \varphi)^*(OO^*)\rangle = \langle X, C_\varphi^{\top}\rangle$$

Main result

Our goal is to characterize $\mathrm{id}_d \otimes \varphi_d(Z_d) \not\succeq 0$ as $d \to \infty$. To understand the meaning of this positivity in the asymptotic sense. Lets first give some definitions

Definition 1. For a ramdom matrix $A_d \in M_n$ with emperical distribution μ_d. We say $\lim_{d\to\infty} A_d$ is *weakly* positive semi-definite if A_d converge to a in distribution and $a \overset{d}{=} \mu \subset [0, \infty)$

Definition 2. For a ramdom matrix $A_d \in M_n$ with emperical distribution μ_d. We say $\lim_{d\to\infty} A_d$ is *strongly* positive semi-definite if A_d converge to a strongly and $a \overset{d}{=} \mu \subset [0, \infty)$

Definition 3. For a ramdom matrix $A_d \in M_n$ with emperical distribution μ_d. We say $\lim_{d\to\infty} A_d$ is *strongly* positive semi-definite if A_d converge to a strongly and $a \overset{d}{=} \mu \subset [0, \infty)$

Lemma 1. Given any matrix $A \in M_{n(\mathbb{C})}$. A is non-positive if and only if there exist a testing vector $v \in \mathbb{C}^n$ with $\|x\| = 1$ such that $x^* A x < 0$

Proof. Suppose $A \not\succ 0$. By definition, there exist a vector $x \in \mathbb{C}^n$ such that $x^* A x < 0$. Consider $x' = \frac{x}{\|x\|}$, we still have $x'^* A x' < 0$. The converse is direct. □

Theorem 2. Given continuous functions f, g. Let $G_d \overset{d}{=} \mu \in M_d \otimes M_n$ be GUE matrix and let $C_{\varphi_d} = f\big(G_d^\top\big), Z_d = g(G_d) \in M_d \otimes M_n$.

Then, for $\varphi_d : M_n \to M_d$ defined by $\varphi_d(X) = (\mathrm{id}_d \otimes \mathrm{Tr}_n)\Big(C_{\varphi_d}(I_d \otimes X^\top)\Big)$

$$\lim_{d\to\infty} \frac{1}{d} \lambda_{\min}((\mathrm{id}_d \otimes \varphi_d)(Z_d)) < 0$$

if and only if

$$\left(\int f g d\mu - \int f d\mu \int g d\mu\right) |\tau(w)|^2 + \left(\int f d\mu \int g d\mu\right) \tau(ww^*) < 0 \quad (1)$$

for some $w \in \mathcal{A}$.

Proof. Observe that for any vector $x \in \mathbb{C}^d \otimes \mathbb{C}^d$. According to Schmidt decomposition, there are two orthornormal basis $\{u_i\}, \{v_i\} \subset \mathbb{C}^d$ such that $x = \frac{1}{\sqrt{d}} \sum_{i=1}^r x_i u_i \otimes v_i$ where $r \le d$ and $x_i > 0$ with $\sum_{i=1}^r |x_i|^2 = \frac{1}{\sqrt{d}}$.

Let $O := \sum_{i=1}^d f_i \otimes f_i \subset \mathbb{C}^d \otimes \mathbb{C}^d$ and U, V be two unitaries such that $U f_i = u_i$ and $V f_i = v_i$ for all $i \in [d]$. Also, let $D = \mathrm{diag}(x_1, x_2, ...x_r, 0, ...0)$ be some diagonal matrix. Then, $x = (UD \otimes V)O$. According to vectorization identity, we have $x = (UDV^\top \otimes I_d)O$. Then, $xx^* = (UDV^\top \otimes I_d)OO^*(UDV^\top \otimes I_d)^*$. It follows that

$$\begin{aligned}
(\mathrm{id}_d \otimes \varphi_d)^*(xx^*) &= (\mathrm{id}_d \otimes \varphi_d^*)(xx^*) \\
&= (\mathrm{id}_d \otimes \varphi_d^*)\Big((UDV^\top \otimes I_d)OO^*(UDV^\top \otimes I_d)^*\Big) \\
&= (UDV^\top \otimes I_n)(\mathrm{id}_d \otimes \varphi_d)^*(OO^*)(UDV^\top \otimes I_n)^* \\
&= (UDV^\top \otimes I_n) C_\varphi^\top (UDV^\top \otimes I_n)^*
\end{aligned}$$

For simplicity, denote $W_d = UDV^\top \otimes I_n \in M_d \otimes M_n$.

$$\begin{aligned}
\langle \mathrm{id}_d \otimes \varphi_d(Z_d), xx^* \rangle &= \langle Z_d, \mathrm{id}_d \otimes \varphi_d^*(xx^*) \rangle \\
&= \langle Z_d, W_d C_\varphi^\top W_d^* \rangle \\
&= n \,\mathrm{tr}\Big(W_d C_\varphi^\top W_d^* Z_d\Big) \\
&= n \,\mathrm{tr}(W_d f(G_d) W_d^* g(G_d))
\end{aligned}$$

Then, since $n > 0$, then we have that $\lim_{d\to\infty} \mathrm{tr}(W_d f(G_d) W_d^* g(G_d)) < 0$ implies $\lim_{d\to\infty} \frac{1}{d} \lambda_{\min}((\mathrm{id}_d \otimes \varphi_d)(Z_d)) < 0$

Lets first assume $W \xrightarrow{\text{dist}} w$. Then, since $\|W\| \leq \|D\| = \max(x_1, x_2, ...x_r) \leq 1$, it is uniformly bounded, acording to Voiculescu's asymptotic freeness, we then have

$$\lim_{d\to\infty} \operatorname{tr}(Wf(G_d)W^* g(G_d)) = \tau(wf(\sigma)w^* g(\sigma))$$
$$= \tau(f(\sigma)g(\sigma))\tau(w)\tau(w^*) + \tau(f(\sigma))\tau(g(\sigma))\tau(ww^*) - \tau(f(\sigma))\tau(g(\sigma))\tau(w)\tau(w^*)$$
$$= (\tau(f(\sigma)g(\sigma)) - \tau(f(\sigma))\tau(g(\sigma)))|\tau(w)|^2 + \tau(f(\sigma))\tau(g(\sigma))\tau(ww^*)$$
$$= \left(\int fgd\mu - \int fd\mu \int gd\mu\right)|\tau(w)|^2 + \left(\int fd\mu \int gd\mu\right)\tau(ww^*)$$

□

Theorem 3. Given continuous functions f, g. Let $G_d \overset{d}{=} \mu \in M_d \otimes M_n$ be GUE matrix and let $C_{\varphi_d} = f\big(G_d^\top\big), Z_d = g(G_d) \in M_d \otimes M_n$.

Then, for $\varphi_d : M_n \to M_d$ defined by $\varphi_d(X) = (\mathrm{id}_d \otimes \mathrm{Tr}_n)\Big(C_{\varphi_d}(I_d \otimes X^\top)\Big)$

$$\lim_{d\to\infty} \lambda_{\min}((\mathrm{id}_d \otimes \varphi_d)(Z_d)) < 0$$

if

$$\gamma \int fgd\mu + (1-\gamma)\int fd\mu \int gd\mu < 0$$

for some $\gamma \in [0, 1]$.

Proof. Observe that for any unit vector $x \in \mathbb{C}^d \otimes \mathbb{C}^d$. According to Schmidt decomposition, there are two orthornormal basis $\{u_i\}, \{v_i\} \subset \mathbb{C}^d$ such that $x = \sum_{i=1}^r x_i u_i \otimes v_i$ where $r \leq d$ and $x_i > 0$ with $\sum_{i=1}^r |x_i|^2 = 1$.

Let $O := \sum_{i=1}^d f_i \otimes f_i \subset \mathbb{C}^d \otimes \mathbb{C}^d$ and U, V be two unitaries such that $Uf_i = u_i$ and $Vf_i = v_i$ for all $i \in [d]$. Also, let $D = \mathrm{diag}(x_1, x_2, ...x_r, 0, ...0)$ be some diagonal matrix. Then, $x = (UD \otimes V)O$. According to vectorization identity, we have $x = (UDV^\top \otimes I_d)O$. Then, $xx^* = (UDV^\top \otimes I_d)OO^*(UDV^\top \otimes I_d)^*$. It follows that

$$\begin{aligned}
(\mathrm{id}_d \otimes \varphi_d)^*(xx^*) &= (\mathrm{id}_d \otimes \varphi_d^*)(xx^*) \\
&= (\mathrm{id}_d \otimes \varphi_d^*)\Big((UDV^\top \otimes I_d)OO^*(UDV^\top \otimes I_d)^*\Big) \\
&= (UDV^\top \otimes I_n)(\mathrm{id}_d \otimes \varphi_d)^*(OO^*)(UDV^\top \otimes I_n)^* \\
&= (UDV^\top \otimes I_n)C_\varphi^\top (UDV^\top \otimes I_n)^*
\end{aligned}$$

For simplicity, denote $W_d = UDV^\top \otimes I_n \in M_d \otimes M_n$.

$$\begin{aligned}\langle \mathrm{id}_d \otimes \varphi_d(Z_d), xx^* \rangle &= \langle Z_d, \mathrm{id}_d \otimes \varphi_d^*(xx^*) \rangle \\ &= \langle Z_d, W_d C_\varphi^\top W_d^* \rangle \\ &= \mathrm{Tr}\Big(W_d C_\varphi^\top W_d^* Z_d\Big) \\ &= \mathrm{Tr}(W_d f(G_d) W_d^* g(G_d))\end{aligned}$$

Then, since we have encounter the form $nd\,\mathrm{tr}(...)$, we can not employ Voiculescu's asymptotic freeness directly. Instead, we will expand this equation directly and do the appropriate approximate. Since G_d is self adjoint, we have $G_d = U_d^* \Lambda_d U_d$ with Harr unitary U. Oberve that $\mathbb{E}_{U_d}[\mathrm{Tr}(W_d U_d^* f(\Lambda_d) U_d W_d^* U_d^* g(\Lambda_d) U_d)]$ which means that $\Phi(W_d^*) = \mathbb{E}_{U_d}[U_d^* f(\Lambda_d) U_d W_d^* U_d^* g(\Lambda_d) U_d]$ is unitarily invariant, that is $\Phi(V^* W_d V) = V \Phi(W_d^*) V^*$. Then, it is well known that such map is of the form [1](eq 7.183)

$$\Phi(W_d^*) = \alpha W_d^* + \beta\,\mathrm{Tr}(W_d^*)$$

we then have

$$\begin{aligned}\mathbb{E}_U[\mathrm{Tr}(W_d f(G_d) W_d^* g(G_d))] &= \mathrm{Tr}(W_d(\alpha W_d^* + \beta\,\mathrm{Tr}(W_d^*))) \\ &= \alpha\,\mathrm{Tr}(W_d W_d^*) + \beta\,\mathrm{Tr}(W_d)\,\mathrm{Tr}(W_d^*) \\ &= \alpha\,\mathrm{Tr}(W_d W_d^*) + \beta |\mathrm{Tr}(W_d)|^2\end{aligned}$$

Now take $W_d = I_{nd}$ and $W_d = \sum_{i,j=1}^{nd} E_{ij}$. Then, we will get

$$\begin{aligned}a_d &= \mathrm{Tr}(f(G_d) g(G_d)) = \alpha(nd) + \beta(nd)^2 \\ b_d &= \mathrm{Tr}(f(G_d))\,\mathrm{Tr}(g(G_d)) = \alpha(nd)^2 + \beta(nd)\end{aligned}$$

solving this gives

$$\alpha = \frac{(nd) b_d - a_d}{nd((nd)^2 - 1)}, \beta = \frac{(nd) a_d - b_d}{nd((nd)^2 - 1)}$$

Note that $\mathrm{Tr}(W_d W_d^*) = \mathrm{Tr}_d(D^2)\,\mathrm{Tr}_n(I_n) = n$ and $\mathrm{Tr}(W_d) = \mathrm{Tr}_d(UDV^\top)\,\mathrm{Tr}_n(I_n) = n\,\mathrm{Tr}(UDV^\top)$ and $\frac{a_d}{nd}$ converge to m_{fg}, $\frac{b_d}{(nd)^2}$ converge to $m_f m_g$ in distribution. Then, putting this back to the equation gives

$$\begin{aligned}&\mathbb{E}_U[\mathrm{Tr}(W_d f(G_d) W_d^* g(G_d))] \\ &= \alpha n + \beta n^2 |\mathrm{Tr}(UDV^*)|^2 \\ &= \frac{n((nd) b_d - a_d)}{nd((nd)^2 - 1)} + \frac{n^2 \left|\mathrm{Tr}_d(UDV^\top)\right|^2 ((nd) a_d - b_d)}{nd((nd)^2 - 1)} \\ &= \frac{a_d}{nd} \frac{n(n^2 d |h|^2 - 1)}{(nd)^2 - 1} + \frac{b_d}{(nd)^2} \frac{n^3 d(d - |h|^2)}{(nd)^2 - 1}\end{aligned}$$

Assume $\frac{|h|^2}{d}$ converge to some γ in distribution. Taking the limit on $d \to \infty$. It becomes

$$\begin{aligned}\frac{a_d}{nd}\frac{n(n^2d|h|^2-1)}{(nd)^2-1}+\frac{b_d}{(nd)^2}\frac{n^3d(d-|h|^2)}{(nd)^2-1} \\ \underset{d\to\infty}{\longrightarrow} m_{fg}n\gamma + m_f m_g(n-n\gamma) \\ = n\big(m_{fg}\gamma + m_f m_g(1-\gamma)\big) \\ = n\left(\gamma\int fgd\mu + (1-\gamma)\int fd\mu\int gd\mu\right)\end{aligned}$$

Now, since $0 \le \lim_{d\to\infty}\frac{\left|\mathrm{Tr}_d(UDV^\top)\right|^2}{d} \le 1$ with the maximum holds when $U = I_d$,$V = I_d$ and $D = \mathrm{diag}(1,0,0,...0)$ and minimum holds when $D = \mathrm{diag}\big(\frac{1}{d},\frac{1}{d},...\frac{1}{d}\big)$. Thus, $\gamma \in [0,1]$.

Then, since n is positive and finite, we obtain that

$$\lim_{d\to\infty}\mathbb{E}_U[\langle \mathrm{id}_d\otimes\varphi_d(Z_d), xx^*\rangle] = n\left(\gamma\int fgd\mu + (1-\gamma)\int fd\mu\int gd\mu\right)$$

We left to show $|\langle \mathrm{id}_d\otimes\varphi_d(Z_d),\, xx^*\rangle - \mathbb{E}_U\langle \mathrm{id}_d\otimes\varphi_d(Z_d),\, xx^*\rangle| \to 0$. It is well known that for L-Lipsthciz function, there are phenomenon of concentration of mass. Specifically, according to [2](Theorem 5.17), it states that (simplified)

Consider $G_{nd} = U(nd)$. Then, if $F : X \to \mathbb{R}$ is L-Lipschitz, and that U is Harr-distributed on G_{nd}. Then for each $t > 0$,

$$\mathbb{P}(F(U) > \mathbb{E}(F(U)) + t) \le \exp\left(-(nd-2)\frac{t^2}{24L^2}\right)$$

We if we consider $F(U) = \mathrm{Tr}(WU^*f(\Lambda)UW^*U^*g(\Lambda)U)$, we will show that it is L-Lipsthciz. This is fairly simple. Given $U, V \in U(nd)$. Using the basic inequality $|\mathrm{Tr}(XY)| \le \|X\|\|Y\|_2$, Observe that

$$\begin{aligned}\|F(U)-F(V)\| \le \|\mathrm{Tr}(W(U-V)^*fUW^*UgU^*)\| + \|\mathrm{Tr}(WV^*f(U-V)W^*UgU^*)\| \\ +\|\mathrm{Tr}(WV^*fVW^*(U-V)^*gU)\| + \|\mathrm{Tr}(WV^*fVW^*V^*g(U-V))\| \\ \le 4\|W\|_2^2\|f(\Lambda)\|\|g(\Lambda)\|\|U-V\|_2 = 4n\|f(\Lambda)\|\|g(\Lambda)\|\|U-V\|_2 \\ \le 4M_d\|U-V\|_2\end{aligned}$$

where $M_d := n\|f(\Lambda)\|\|g(\Lambda)\|$, then

$$\mathbb{P}(F(U) > \mathbb{E}F(U) + t) \leq \exp\left(-(nd-2)\frac{t^2}{384M_d^2}\right) = \exp\left(-\frac{nd}{384M_d^2} + \frac{t^2}{384M_d^2}\right)$$
$$= c\exp\left(-\frac{nd}{384M_d^2}\right)$$

Since GUE has spectrum converging to $[-2,2]$ strongly, $\|f(\Lambda)\| \leq \sup_{x\in[-2,2]}|f(x)| \leq R$ and similar for $\|g(\Lambda)\| \leq R$ as $d \to \infty$. We have $M_d \leq R^2$ uniformly as $d \to \infty$. It then follows that $c\sum_{d=1}^{\infty}\exp\left(-\frac{nd}{384M_d^2}\right) < \infty$. According to Borel Cantelli theorem, we have

$$\mathbb{P}\left(\limsup_{d\to\infty}\{F(U) - \mathbb{E}F(U) > t\}\right) = 0$$

that is

$$F(U) \to \mathbb{E}F(U) \quad a.s.$$

The results follows. □

Bibliography

[1] J. Watrous, *The Theory of Quantum Information*. Cambridge University Press., 2018.

[2] E. Meckes, *The Random Matrix Theory of the Classical Compact Groups*. 2019.